



UHP Stainless Steel Best Known Methods (BKM) Guidelines

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Global Facilities - Design & Construction Standard - Gas - UHP Stainless Steel Guidelines

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Item	Description
1 General Information	
1.1 Document Name	Ultra High Purity (UHP) Stainless Steel best known method
1.2 Document Number	CSG-3
1.3 Description	Material specifications, manufacturing, cleaning, packaging, and installation requirements for ultra high purity (UHP) stainless steel materials and components utilized to supply and distribute fluids in the semiconductor manufacturing process.
1.4 Owner	stamenghi
1.5 Applicable FPM System IDs	CSG/BSG, PC
1.6 Revision History	1
1.7 Glossary of Terms	http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef83a1f41c
2 Minimum Recommendations	Hide or Display Rows as Needed
2.1 Mechanical	

2.1.1 Design Parameters	1. Stainless steel tubing, materials, and components to supply and distribute process fluids for the manufacturing process. Ultra high purity (UHP) stainless steel materials shall meet, but not be limited to, the following requirements: 1.1 ASTM A269, A632, and A312 standards 1.2 Tubing and butt weld fittings hardness shall be between 60 and 90 on the Rockwell b (Rb) scale. 1.3 Proof pressure shall meet design pressure with 20% over capacity as a minimum. 1.4 Seamless tubing shall be utilized for sizes ≤ 1 inch 1.5 Seamless or welded tubing shall be utilized for sizes > 1 inch ≤ 2 inches according to design standard, manufacturer capability and/or directed in the design phase. 1.6 Welded tubing shall be utilized for sizes > 2 inches
2.1.2 Operating Parameters	
2.1.3 Design Configuration	
2.1.4 Capacity	
2.1.5 Redundancy	
2.1.6 Expansion	
2.1.7 Operation	1. Field bending of tubing is allowed for line sizes up to 1 inch, with minimum 10x-tube diameter radius bends. 2. Field bending of 1/4 and 3/8 inch size tubing may be allowed using lower radius bending than 10x diameter, as specified in site construction specifications. Bending execution shall be further detailed in the site specifications to prevent tubing stress with loss of electro-polishing and passivated layer. Bending should be executed as slowly as possible to allow layer to change and adapt its shape to the new profile, minimizing the layer distortion and breakage. 3. Coaxial tubing for process pipeline must be bent so that the inner tube meets the above requirements.
2.1.8 Maintenance	

Item	Description
2.1.9 Materials	1. All UHP stainless steel tubing shall be manufactured with 316L stainless steel single melt as a minimum, and electro-polished with 10μinch Ra (0.25μm) average and 20μinch Ra (0.50μm maximum internal finishing. 2. All UHP stainless steel components, such as valves, regulators etc, shall be manufactured with: 2.1 316L stainless steel single melt as a minimum, for non HPM, inert and asphyxiant gases, and electro-polished with 10μinch Ra (0.25μm) average and 20μinch Ra (0.50μm maximum internal finishing. Remarks: For easier and safer management of large installation projects by contractors, it is recommended using these single melt electro-polished stainless steel component for non HPM bulk gas only. 2.2 316L stainless steel double melt (vacuum arc remelted) as a minimum, for HPM, toxic, corrosive and oxidizer gases, and electro-polished with 10μinch Ra (0.25μm) average and 20μinch Ra (0.50μm maximum internal finishing. 3. All UHP stainless steel wetted parts shall be manufactured and followed by oxidization (passivated) to remove free iron and produce chromium, nickel, iron, and molybdenum oxides in the stainless steel layer. Tubing and weld fittings shall be annealed. 4. Metal seal gaskets for vacuum coupling radiation laboratory connections (VCR) shall be made of electro-polished stainless steel or nickel for all compatible applications. Gasket retainers shall be used when required by installation, generally in horizontal coupling, and site design construction procedures. 5. Carbon Monoxide (CO) is not compatible with nickel gaskets due to reaction with nickel producing Nickel Penta-Carbonyl, as process contaminant, and requires stainless steel VCR gaskets. Similarly, for the same reason filters elements shall be made of Stainless Steel.
2.1.10 Components	1. All metal wetted parts of UHP valves, pressure regulators, filter housings, pressure transducers, and other components shall conform to requirements indicated in this BKM except for those applications where other materials are required for chemical compatibility and when requirements are clearly identified for specific materials. For example, diaphragms and poppets in regulators utilized in corrosive fluid media such as HBr, HCl, and others shall have Hastelloy C-22 trim, depending on vendor recommendations and engineering and construction standards. 2. Valves to be installed on non-HPM gas distributions shall have downstream purge ports, as a minimum, to allow for specific purging and venting during construction and qualification. Other requirements shall be identified in the design phase according to specific application.

2.1.11 Manufacturers	1. Tubing manufacturers: Valex, Cardinal, Dockweiler, Swagelok, Sumikin, Kuze, Plymouth 2. Valves manufacturers: Swagelok, APTech, Parker, Fujikin, Carten
2.1.12 Installation	1. Tubing and other components shall be welded using an orbital welding process. Welding shall be performed using 0.003 μ filtered argon to assure inert environment. 2. Argon purity for welding shall have 99.995% purity as a minimum. 3. Welding and welder's certification shall be provided according to local codes and regulations, and site specification. 4. Continuous nitrogen or other inert gas flow shall be maintained in piping system segments during cutting, facing, fabrication, and installation. Pulse purge shall be executed after cutting and facing to remove metal scraps and particles. 5. VCR connections shall be utilized as a coupling method in all bulk and specialty gas systems, liquid chemical dispensing and distribution such as TEOS and others precursors. VCR shall be used when the sequence of installed components does not allow welding, when replacement of components is required for maintenance and/or system upgrade and when defined by system design and site construction specifications. 6. Mechanical compression fittings are allowed for organic solvent dispense and distribution systems, such as IPA, PGMEA systems, and others based on engineering standard, site construction specifications, and directed in the design phase.

Item	Description
2.1.13 QA/Testing	1. Stainless steel materials and components shall be manufactured and finally cleaned to meet high purity requirements. Manufacturers procedures shall meet or exceed cleanliness based on, but not be limited to, the following procedures: 1.1 Rinsed with 0.1 μ filtered, 18 megohm ultra pure water (UPW) and heated to 140°F (60°C) until the effluent meets manufacturer specification. 1.2 Dried with 0.01 μ filtered nitrogen 2. Final materials and components assembling and packaging shall be executed in an ISO Class 6 clean room space. Tubing shall be capped; other components shall be appropriately capped or plugged depending on the port shape and configuration. Materials and components shall be bagged and purged with 0.01 μ filtered nitrogen or argon for shipment and storage. 3. Welding of tubing and components shall be 100% tested by Micron, a contractor, or a third party qualified in Quality Control (QC). Testing, inspection, and acceptance criteria shall be identified in each site construction specification according to local codes and regulations. 4. Coupons, welding tests, and programs shall be kept for tracking based on each QC program, as defined in each site construction specification. 5. Semi standard for material specification, machining and welding techniques, internal finishing process and quality inspection are reported as reference in the sections 6.2.17 to 6.2.23.
2.1.14 Other	1. SEMI standard for reference in the section 6.3.
2.2 Electrical	
2.3 I & C	
2.4 Chemical	
2.4.1 Design Parameters	1. This document applies to all materials and components utilized in the organic solvents dispense and distribution systems such as IPA, PGMEA HMDS, and others, except for specific non-compatible products and applications. Exclusions of Stainless Steel utilization shall be identified in the Engineering standards, and/or directed in the design phase, 2. Organic solvents supply pressure and temperature is identified in the "UHP Chemical Dispense BKM's" (Active link in section 6.5.3). 3. Chromium content in the 316L stainless steel shall be between 16% and 18%. 4. Sulfur content in seamless tubing shall be between 0.005% and 0.012%. 5. Sulfur content in welded tubing shall be between 0.005% and 0.017%. 6. Butt weld fittings and components may have sulfur content lower than 0.005% according to manufacturer standards, however, the welding program must be re-validated, coupons must be welded and inspected according to the QC acceptance criteria before installing these micro fittings in UHP SS systems.
2.4.14 Other	1. SEMI standard reference. 1.1 SEMI E49.4. Guide for high purity solvent distribution systems in semiconductor manufacturing equipment.
2.5 Gas	

2.5.1 Design Parameters	1. This document applies to all materials and components utilized to supply and distribute fluids to the manufacturing process by stainless steel systems. These fluids include bulk gases, bulk and cylinder specialty gases, organic liquid chemicals, and precursors based on, but not limited to, the following: TEOS, TEPO, TEB, TiCl₄, TMA, TDMAH, TEMAZ, Pyridine, Hexachloro-Disilane (HCDS) and others. 2. Specialty gas supply pressure and temperature is identified in the "Specialty Gas Supply Systems BKM Guidelines" (Active link in section 6.5.1). 3. Some of the fluids utilized in the manufacturing process may need specific temperature control and/or may develop temperature change due to their physical properties from temperature identified in BKM Standards parameters guidelines.
2.5.9 Materials	1. See linked procedures and/or procedures/specification in section 6.2 as reference.
2.5.10 Components	1. See linked procedures and/or procedures/specification in section 6.2 as reference.
2.5.12 Installation	1. See linked procedures and/or procedures/specification in section 6.2 as reference. 2. Some fluorine mixtures and fluoride molecules needs to have their gas manifolds and pipelines passivated. Detailed guidance is specified in the "Specialty gas distribution BKMs guidelines" (Active link in section 6.5.2). 3. VCR locking systems shall be utilized to avoid radial torque on the VCR connection with potential leak, based on, but not limited to, the following situations: 3.1 Where clearly identified in the design documents. 3.2 Any other specific application identified in the site construction specifications.

Item	Description
2.5.13 QA/Testing	1. Moisture tests shall be performed in all new installations to ensure dry down as part of qualification tests. Maximum moisture concentration $\leq 40\text{ppb}$. 2. Oxygen tests shall not be routinely performed as it is assumed it follows the same decay trend of moisture. Oxygen test requirements shall be identified in the site construction specifications and directed during the design phase. 3. Particle contamination shall be controlled to ensure that pipelines, manifolds, and assemblies have been appropriately built, installed, and pre-cleaned. A shock purge procedure shall be performed to increase particle removal efficiency as an auxiliary procedure for final qualification and shall be defined in each site construction specification. 4. Particles shall be measured by: 4.1 Laser particle counter (LPC) 4.2 Condensation nucleus particle counter (CNC) 5. Particle measurement shall be executed based on, but not limited to, the following methods: 5.1 Steady flow measurements. 5.1.1. Laser Particle Counter (LPC) 5.1.1.1 Analyze 1cft per run. Three consecutive runs, as a minimum. It is recommended to utilize particle counters running at 1cft/min to minimize qualification time. 5.1.1.2 ≤ 3 counts of 0.2μ and larger particles 5.1.2. Condensation Nucleus Counter (CNC) 5.1.2.1 Analyze 1cft per run. Three consecutive runs, as a minimum. 5.1.2.2 ≤ 3 counts of 0.01μ and larger particles 5.2 Pulse Purge Measurement. 5.2.1. Laser Particle Counter (LPC) 5.2.1.1 ≤ 10 counts of 0.2μ and larger particles for 5 consecutive 1-minute sampling periods. 6. Site procedures and specifications may contain additional requirements based on a different scope of qualification, i.e., new single component for site acceptance criteria and contractor/third party specifications. Site procedures and specifications are linked in section 6.2 as a reference.

2.5.13 QA/Testing – continued	7. Semi standard for quality controls, assurance, and test methods inspection are reported as referenced in sections 6.3.1 to 6.3.14. 8. Leak tests and/or pressure tests, moisture tests, and particle tests shall be executed to guarantee integrity and quality of installations before releasing for process fluids. 9. Leak tests shall be executed according to SEMI F1 and local codes and regulations (i.e., ASME B31.3 345.1) as follows: 9.1 Inboard helium leak check: Pull high vacuum into the line to be tested, spray helium close and across welded and mechanical connections. Leak rates not to exceed 1×10^{-9} Atm cc/sec. 9.2 Outboard leak check: Fill the line to be tested with a helium or helium mixture. Sniff each weld and mechanical connection. Readings above 1×10^{-6} Atm cc/sec should be considered a leak. 10. If pressure testing is required, it shall be performed as directed by site construction specifications. 11. Each site shall periodically re-evaluate their specifications to determine the most appropriate and cost-effective methods.
2.6 Water & Waste Water	
2.7 Other	
3 Best Practices	Hide or Display Rows as Needed
3.1 Mechanical	
3.2 Electrical	
3.3 I & C	
3.4 Chemical	
3.5 Gas	
3.6 Waste & Waste Water	
3.7 Other	
4 Supporting Documentation	
4.1 Sequence of Operations	
4.2 System Acceptance Criteria	
4.3 Design Concept	
4.4 P & IDs	
4.5 Safety	
4.6 Block Diagrams	
4.7 Other Technical Document	
5 Approved Variances	[Place holder – Add link to variances]
5.1	
5.2	

Item	Description
5.3	
6 Other Links	
6.1 Definitions	
6.2 References	1. F6 high purity stainless steel tubing systems specification – http://edm.micron.com/cgibin/mtgetdoc.exe?itemID=090007da80053add 2. F6 High purity installation procedure – http://edm.micron.com/cgibin/mtgetdoc.exe?itemID=090007da8002ece8 3. F4 SECTION 15240A HIGH PURITY STAINLESS STEEL PIPING SYSTEMS – http://edm.micron.com/cgibin/mtgetdoc.exe?itemID=09005aef800a7bd0

6.3 Industry Standards	1. SEMI F20-706E. Specification for 316L SS bar, forging, extruded shapes, plate, and tubing for components used in general purpose, high purity, and ultra-high purity semiconductor manufacturing applications 2. SEMI F16. Specification for 316L stainless steel tubing which is to be finished and electro-polished for high purity semiconductor manufacturing applications. 3. SEMI F19. Specification for the surface condition of the wetted surfaces of SS components 4. SEMI F1. Specification for leak integrity of high purity gas piping systems and components 5. SEMI F44. Specifications for machined SS weld fittings 6. SEMI F45. Specification for machined SS reducing weld fittings 7. SEMI F78. Practice for gas tungsten arc (GTA) welding of fluid distribution systems in semiconductor manufacturing application 8. SEMI F81. Specification for visual inspection and acceptance of gas tungsten arc (GTA) welds in fluid distribution systems used in semiconductor manufacturing applications 9. SEMI F106. Test method for determination of leak integrity of gas delivery systems by Helium leak detector. 10. SEMI E49.8. Guide for high purity and ultra high purity gas distribution systems in semiconductor manufacturing equipment 11. SEMI F37. Method for determining surface roughness parameters for gas distribution system components 12. SEMI F60. Test method for ESCA evaluation of surface composition of wetted surfaces of passivated 316L components. 13. SEMI F72. Test method for AUGER electron spectroscopy (AES) evaluation of oxide layer of wetted surfaces of passivated 316L components. 14. SEMI F73. Test method for scanning electron microscopy (SEM) evaluation of wetted surfaces condition of SS components.									
6.4 Procedures										
6.5 Interfacing System Standards	1. Specialty Gas Supply Systems BKM Guidelines.pdf http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef86707a34 2. Specialty Gas Distribution Systems BKM Guidelines.pdf http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef8673b75b 3. UHP Chemicals dispense BKMs – http://edm.micron.com/cgibin/mtgetdoc.exe?itemID=09005aef833fe0c9									
7 Notes										
8 Revision History										
0	9/25/2015 Initial issue									
Admin	2/1/2017 Change document number CSG-2 to CSG-3 (per Stefano)									
1	5/18/2017 Update of linked procedures, SEMI standard reference, and other minor changes (Stefano).									
9 Appendix										
	<table><tr><th>Term/Acronym</th><th>Definition / Description</th></tr><tr><td>FCT</td><td>Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs</td></tr><tr><td>GFTT</td><td>Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards</td></tr><tr><td>GFC</td><td>Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide</td></tr></table>	Term/Acronym	Definition / Description	FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs	GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards	GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide	
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		Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
	Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
	Global Facilities Construction Team	This is used interchangeably with Global Construction Team
	Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
	Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team